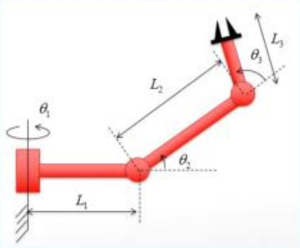
Inverse Kinematics Tutorial

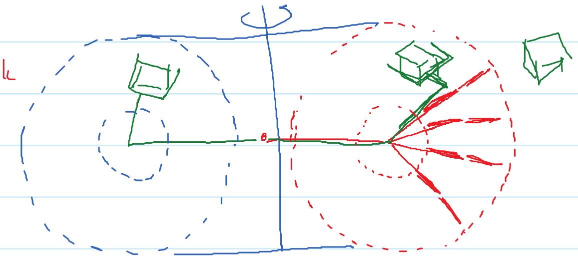
26 March 2021 13:47

Question 1

- Sketch the workspace of the following manipulator with:
  - L1 = 15
  - L2 = 10
  - L3 = 3



for θ_1 = $-180°$ to $180°$
for θ_2 = $-180°$ to $180°$
for θ_3 = $-180°$ to $180°$
calculate ${}^0P$ (end effector)
end
end
end
plot all the ${}^0P$



Question 2

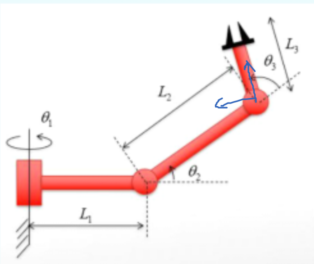
- The last frame of the 3-link manipulator on the right is required to be at:

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

want

match

- What should the joint angles be?



General expression for ${}^0_3T$
↳ function of $\theta_1, \theta_2, \theta_3$

from forward kinematics tutorial

$${}^0_3T = \begin{bmatrix} C_1C_{23} & -C_1S_{23} & S_1 & L_1C_1 + L_2C_1C_2 \\ S_1C_{23} & -S_1S_{23} & -C_1 & L_1S_1 + L_2S_1S_2 \\ S_{23} & C_{23} & 0 & L_2S_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

for θ_1 :

$$S_1 = r_{13}$$

$$C_1 = -r_{23}$$

$\rightarrow$ $\cancel{\theta_1 = \arcsin(r_{13})}$

$$\boxed{\theta_1 = \arctan2(r_{13}, -r_{23})}$$

for θ_2 :

$$p_z = L_2S_2$$

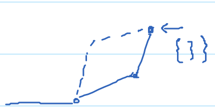
$$S_2 = \frac{p_z}{L_2}$$

$\longrightarrow$ $\cancel{\theta_2 = \arcsin\left(\frac{p_z}{L_2}\right)}$

$$C_2 = \pm\sqrt{1 - S_2^2} = \pm\sqrt{1 - \frac{p_z^2}{L_2^2}}$$

$\Rightarrow$

$$\boxed{\begin{aligned}\theta_2 &= \arctan2(S_2, C_2) \\ &= \arctan2\left(\frac{p_z}{L_2}, \pm\sqrt{1 - \frac{p_z^2}{L_2^2}}\right)\end{aligned}}$$



2 solutions

for θ_3:

$$\left.\begin{aligned} S_{23} &= r_{31} \\ C_{23} &= r_{32} \end{aligned}\right\} \quad \theta_2 + \theta_3 = \operatorname{arctan2}(r_{31}, r_{32})$$

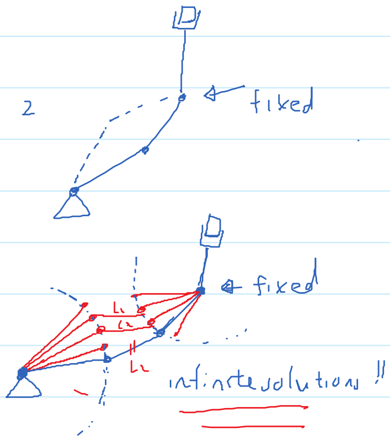
$\theta_2 + \theta_3$

$$\boxed{\theta_3 = \operatorname{arctan2}(r_{31}, r_{32}) - \theta_2}$$

2 solutions

Question 3

- Given a desired position and orientation of the hand of a three-link planar rotary jointed manipulator,
  - there are two possible solutions.
  - If one more rotational joints is added (while the arm is still planar),
  - how many solutions are there?

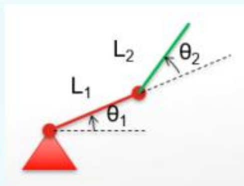


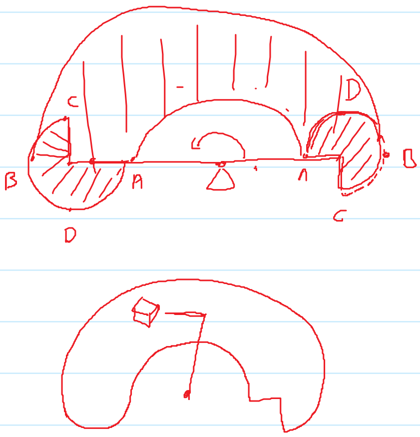
Question 4

- For the two-link robot as shown, the second link is half as long as the first ($L_1 = 2L_2$).
  - The joint range limits are:

$$0 < \theta_1 < 180$$
$$-90 < \theta_2 < 180$$

  - Sketch the approximate reachable workspace of the tip of link 2.



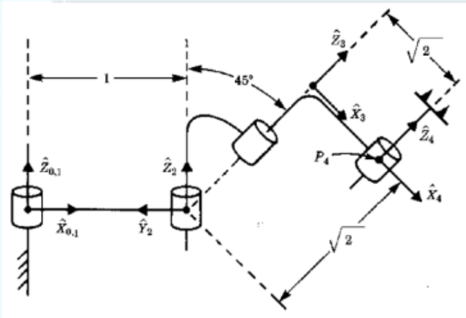


Question 5

- For the 4R manipulator as shown, the non-zero link parameters are:

  $a_1 = 1, \alpha_2 = 45°, d_3 = \sqrt{2}, a_3 = \sqrt{2}$
- The configuration as shown is:

  $\theta_1 = 0, \theta_2 = 90°, \theta_3 = -90°, \theta_4 = 0°$
- Each joint has limit of +/- 180°.
- Find all values of θ_3 such that: ${}^0P_{4ORG} = [1.1 \quad 1.5 \quad 1.707]^T$.



General ${}^0_4T$

DH - Table

i	a_{i-1}	α_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	1	0	0	θ_2
3	0	45°	$\sqrt{2}$	θ_3
4	$\sqrt{2}$	0	0	θ_4

${}^0_1T, {}^1_2T, {}^2_3T, {}^3_4T \rightarrow {}^0_4T$

$$
{}^{i-1}_{\ \ i}T = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 1 \\ s\theta_2 & c\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & 0 \\ 0.707\, s\theta_3 & 0.707\, c\theta_3 & -0.707 & -0.707\sqrt{2} \\ 0.707\, s\theta_3 & 0.707\, c\theta_3 & 0.707 & 0.707\sqrt{2} \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^3_4T = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & \sqrt{2} \\ s\theta_4 & c\theta_4 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

$$
{}^0_4T = {}^0_1T \cdot {}^1_2T \cdot {}^2_3T \cdot {}^3_4T
$$

$= \ldots\ldots$

$$
= \begin{bmatrix} * & * & * & \sqrt{2}\, c_{12} c_3 - \overbrace{\sqrt{2} \cdot 0.707}^{1}\, s_{12} s_3 + \sqrt{2} \cdot 0.707\, s_{12} + c_1 \\ * & * & * & \sqrt{2}\, s_{12} c_3 + \sqrt{2} \cdot 0.707\, c_{12} s_3 - \sqrt{2}\, 0.707\, c_{12} + s_1 \\ * & * & * & \boxed{\underbrace{\sqrt{2} \cdot 0.707}_{1}\, s_3 + \underbrace{0.707\sqrt{2}}_{1}} \\ 0 & 0 & 0 & \end{bmatrix}
$$

want

$$
= \begin{bmatrix} * & * & * & 1.1 \\ * & * & * & 1.5 \\ * & * & * & \boxed{1.707} \\ 0 & 0 & 0 & 1 \end{bmatrix}
$$

θ_3 ☞ $\quad s_3 + 1 = 1.707$

$s_3 = 0.707 \quad \rightarrow \quad \cancel{\theta_3 = \text{atan}(0.707)}$

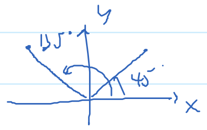
$c_3 = \pm\sqrt{1 - s_3^2}$

$= \pm\sqrt{1 - 0.707^2}$

$= \pm\, 0.707$

$$\theta_3 = \operatorname{arctan2}(s_3, c_3) = \operatorname{arctan2}(\underbrace{0.707}_{y}, \underbrace{\pm 0.707}_{x})$$

$$= 45^\circ, 135^\circ$$



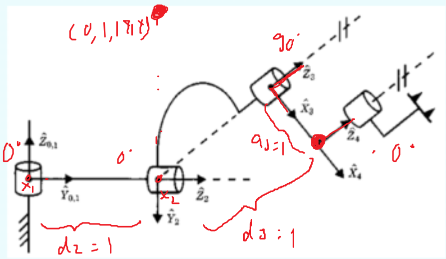
Question 6

d_2 = distance from X_1 to X_2 along Z_2

d_3 = " X_2 to X_3 " Z_3

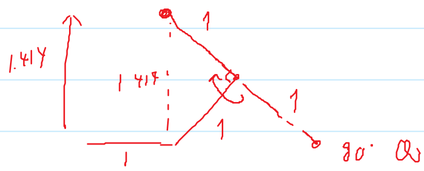
a_3 = " Z_3 to Z_4 " X_3

- For the 4R manipulator as shown, the non-zero link parameters are: $\alpha_1 = -90°, \alpha_2 = 45°, d_2 = 1, d_3 = 1, a_3 = 1$.
- The configuration as shown is: $\theta_1 = 0, \theta_2 = 0°, \theta_3 = 90°, \theta_4 = 0°$
- Each joint has limit of +/- 180°.
- Find all values of θ_3 such that: ${}^0P_{4ORG} = [0 \quad 1 \quad 1.414]^T$.



not to scale

$\theta_3 = -90°$



$$\boxed{\theta_3 = -90°}$$